chemosensors

Review

Electrochemical Sensors and Biosensors for the Detection of Pharmaceutical Contaminants in Natural Waters—A Comprehensive Review

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Abstract: This review highlights recent advances in the development of electrochemical sensors and biosensors for detecting pharmaceutical contaminants in water samples, including surface water, wastewater, and bottled water. The electrode modification materials—such as nanomaterials, conductive polymers, and eco-friendly nanocomposites—have shown notable improvements in sensor sensitivity and selectivity, enabling the detection of various pharmaceutical compounds, including diclofenac, ibuprofen, antibiotics, and hormones. Future research directions suggest the use of eco-friendly nanocomposites, the development of multiplexed platforms for simultaneous contaminant detection, and the integration of sensors into portable devices for in situ monitoring. Additionally, integrating biosensors with specific biomolecules and IoT technology can expand their applicability, enabling continuous and efficient water quality monitoring, thus supporting public health and environmental protection.

Keywords: water quality; drugs; environment; health; nanomaterials

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1. Introduction

The presence of pharmaceutical contaminants, considered emerging pollutants in natural waters, is a significant public health and environmental concern. The frequent and often inappropriate use of pharmaceutical products, both for human and veterinary purposes, contributes to this issue [1]. Rural and urban communities, which rely primarily on surface and groundwater, are highly exposed to a wide range of pharmaceuticals present in water. These residues originate from various sources, such as hospital effluents, industrial discharges, and domestic wastewater, and include a diverse array of substances: antibiotics, analgesics, antidepressants, hormones, and other pharmaceuticals [2,3]. Most of these substances are not completely removed by conventional water treatment processes, and their presence in surface waters can have adverse effects on aquatic ecosystems and human health [4,5]. Among the frequently detected pharmaceuticals in river waters are antibiotics such as fluoroquinolones or beta-lactams, which enter the water through industrial effluents,

hospital residues, and improper disposal of medicines [6–8]. Antibiotics present in water can lead to the development of resistant bacteria and resistance genes, contributing to aquatic environmental pollution, affecting water quality and ecosystems, and posing a risk to human health [9,10]. Analgesics and anti-inflammatories such as ibuprofen or diclofenac, excreted by humans and released through industrial discharges, can cause significant toxicity to phytoplankton, algae, and bacteria, with harmful effects on kidneys, gills, and other organs in freshwater fish, adversely impacting bone development, the immune system, and potentially inducing genotoxicity [11–14]. Psychiatric medications, such as carbamazepine and venlafaxine, are difficult to remove from wastewater, and their bioaccumulation can contaminate drinking water and the natural environment, leading to neurological, endocrine, reproductive, and metabolic disorders [15–17]. Hormones from contraceptives and other hormonal therapies entering waters can disrupt endocrine systems in aquatic organisms, affecting reproduction and population balance and pose a risk to human health through contamination of the food chain [18].

Water contamination with pharmaceuticals is a global issue that varies significantly between developed and developing countries, influenced by water treatment infrastructure, population density, and economic activities [19]. In developed countries such as Germany, France, and Finland, carbamazepine is the most frequently detected pharmaceutical in surface waters, with high concentrations linked to the extensive use of this medication. Although advanced, conventional treatment systems are not fully effective in removing pharmaceuticals [20].

In developing countries such as India, treatment systems are even less efficient, resulting in the presence of a wide range of pharmaceuticals, including antibiotics and anti-inflammatory drugs, in rivers [21]. In Vietnam, antibiotics are detected in both river water and sediments, reflecting their extensive use in medicine and agriculture [22]. Studies from Japan and Taiwan confirm the presence of pharmaceutical compounds in rivers and groundwater, with population density playing a significant role in contamination [23].

Seasonal differences also influence contamination levels. For instance, in the Chaobai River in China, concentrations are higher during winter when natural dilution is reduced. Additionally, the use of reclaimed water for irrigation contributes to groundwater contamination, a phenomenon more pronounced in developing regions [24].

Rapid and accurate detection of pharmaceutical contaminants in river waters is essential to protect the environment and public health. Effective monitoring allows risk assessment, implementation of preventive and control measures, and optimization of water treatment processes. In this context, detection technologies must be not only sensitive and selective but also adaptable for use under diverse environmental conditions. Traditional analytical methods, such as liquid chromatography–mass spectrometry [25], gas chromatography–mass spectrometry [26], high-performance liquid chromatography [27], UV-Vis spectroscopy [28], or capillary electrophoresis [29], although remarkably precise, are often expensive, labor-intensive, and require sophisticated equipment, limiting their applicability for real-time and on-site monitoring. In contrast, electrochemical methods are not only more accessible and easier to use in such conditions, but they can often provide more efficient detection limits than conventional methods in certain applications. For example, for citalopram, the HPLC-DAD method following magnetic solid-phase extraction (MSPE) achieved a LOD of 1.43 ng/mL (4.41 μM) [30], compared with the developed electrochemical sensor, which achieved a much lower LOD of 0.041 μM [31]. In another study, using HPLC with UV-fluorescence, a less efficient detection limit (60 ng/L) for ibuprofen [32] detection was obtained, compared with differential pulse voltammetry, which provided a much lower detection limit (0.0005 nM = 0.103 ng/L) [33].In this context, electrochemical sensors and biosensors have proven to be promising solutions for moni-

toring pharmaceutical contaminants in river waters. For example, electrochemical sensors based on screen-printed electrodes (SPE) [34], glassy carbon (GCE) [35], or carbon paste (CPE) [36] have been used for the rapid detection of pharmaceutical compounds in water. Additionally, electrochemical biosensors, which incorporate enzymes or antibodies, can detect specific pharmaceutical contaminants with higher accuracy and selectivity compared with sensors [37,38]. These devices enable the efficient identification of pharmaceutical substances, thus contributing to water quality monitoring and management. The fundamental differences between electrochemical sensors and biosensors lie in the nature of the recognition mechanism: sensors rely on generally non-specific chemical or physical interactions at the electrode surface, while biosensors employ specific molecular recognition mechanisms because of integrated biological components. While electrochemical sensors can be more stable and less susceptible to degradation, biosensors offer superior sensitivity and specificity, being capable of detecting contaminants even in extremely low concentrations [39].

This paper provides a detailed evaluation of electrochemical sensors and biosensors used for detecting pharmaceutical contaminants in various water samples (surface, wastewater, or bottled water), focusing on their fundamental characteristics, such as sensitivity, selectivity, and detection limits. The advantages and limitations of sensor-modifying elements are examined, emphasizing their performance under specific environmental conditions and their capacity to detect pharmaceutical contaminants. This study also offers a comparative presentation of electrochemical sensors and biosensors, highlighting recent innovations and technological advancements over the last 10 years, such as the integration of nanostructured materials or advanced biological elements, and explores the impact of these improvements on real-time pollutant monitoring capabilities. Future research directions are identified by analyzing the limitations in sensor or biosensor construction, aiming to optimize them for water monitoring applications and other environmental samples. Additionally, the practical applicability of these devices in water quality monitoring programs is discussed, considering factors such as long-term stability, production and maintenance costs, as well as miniaturization possibilities and integration into complex monitoring systems.

Unlike previous research, which generally reported on the detection of a single compound [39] or a specific class of pharmaceutical substances [34,40] or exclusively utilized sensors [41] or biosensors [42,43], this study proposes a multidimensional approach that integrates both electrochemical sensors and biosensors for detecting a wide range of pharmaceutical substances from different classes in natural water samples. Furthermore, in contrast to other studies that focus more on the fundamental technology behind sensor construction [44], this study highlights the latest innovations in the development of modifying materials and functionalization methods that enhance the performance and application diversity of these devices. The integration of sensors and biosensors, together with the use of advanced materials, brings added value and uniqueness compared with existing research in the field.

The challenges associated with electrochemical sensors and biosensors are addressed, including their performance under varying environmental conditions, long-term stability, production and maintenance costs, as well as their integration into complex monitoring systems. The impact of these advancements on real-time pollutant monitoring is explored, emphasizing the importance of innovations in optimizing performance for advanced water analysis and other environmental sample applications.

2. Pharmaceutical Contaminants in River Waters and Modern Removal Methods

2.1. Legislative Regulations

The contamination of surface waters with pharmaceutical substances is an emerging environmental issue that has drawn the attention of the international community, prompting governments and international organizations to begin developing specific regulations.

A global study [45] conducted in 1052 locations across 104 countries on all continents revealed that pharmaceutical pollution in rivers is a major concern, with concentrations of pharmacologically active substances exceeding safe levels in over a quarter of sites. The highest average cumulative pharmaceutical concentrations were found in Lahore, Pakistan (70.8 µg/L), La Paz (68.9 µg/L), and Addis Ababa (51.3 µg/L), with the most polluted site being Rio Seke in La Paz, measuring 297 µg/L due to untreated wastewater discharge and medical/pharmaceutical waste. Cumulative pharmaceutical concentrations were calculated at each sampling site as the sum of all quantified active pharmaceutical ingredients at that specific location. Out of the sixty-one targeted substances, fifty-three were detected in at least one location (Figure 1), with caffeine, nicotine, acetaminophen, and cotinine found on every continent. An additional 14 substances were detected on all continents except Antarctica. The highest concentrations were for paracetamol, caffeine, metformin, fexofenadine, sulfamethoxazole, and gabapentin, and concentration variability may be influenced by regional differences in accessibility and regulatory frameworks for medications.

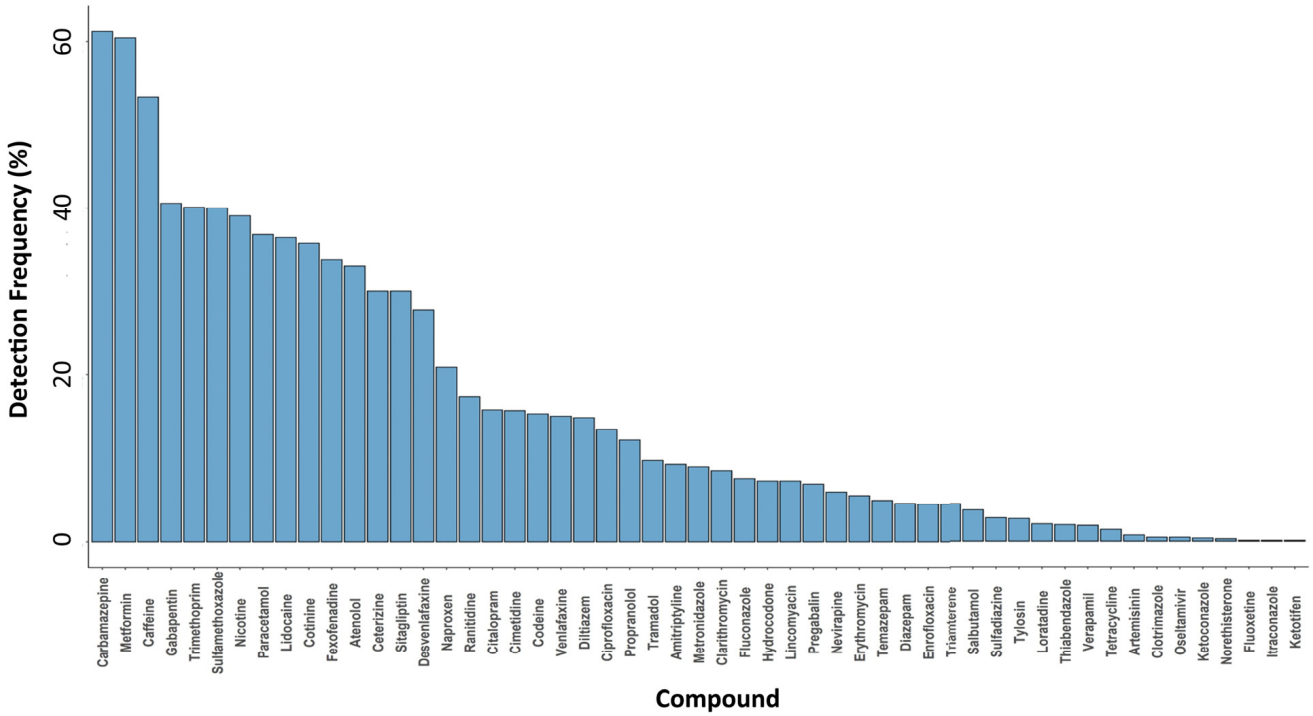


Figure 1. Pharmaceutical compounds detection frequency [45].

In many countries around the world, current legislation regarding pharmaceutical pollution remains insufficient and fragmented. The spread of pharmaceutical substances in aquatic environments varies significantly depending on their usage and the physicochemical properties of the active compounds (pKa and solubility), as well as the characteristics of wastewater (acidity, turbidity, and dissolved oxygen) [46]. The concentrations of pharmaceutical substances or those from cosmetic products in groundwater and surface waters are typically below 100 ng/L, while in treated water, they are less than 50 ng/L [47]. The first

evidence of the presence of pharmaceuticals in the aquatic environment was reported in the United States in 1976 [47]. Subsequent studies have identified numerous pharmaceutical substances in various regions, varying according to consumption habits and environmental conditions. Germany's Ministry of the Environment identified 631 pharmaceutical substances in aquatic environments across 71 countries, with significant detections in the USA, Europe, Asia, and other regions [48]. Other studies reported the presence of approximately 203 pharmaceutical contaminants in 41 countries [49], and studies suggest that detection of these contaminants is more frequent in developed countries, while studies remain limited in developing regions such as Africa [46,50–52].

Romania incorporates European environmental protection legislation through Decision No. 188/2002, which regulates the standards for collecting, treating, and discharging wastewater into aquatic environments. This document establishes strict advanced treatment requirements for urban areas with over 10,000 population equivalents, given that the entire territory of Romania has been designated as a sensitive area in the context of protecting the Danube basin and the Black Sea [53].

Regulations regarding pharmaceutical substances in the environment vary based on the intensity of control measures and can be classified into three levels [54].

Level 1 involves comprehensive measures with strict legal regulations to reduce pharmaceutical residues. California, for example, mandates the monitoring of pharmaceutical substances in water reuse projects [55], and Switzerland has stringent regulations to reduce micropollutants [56].

Level 2 entails moderate interventions based on global or local guidelines without strict legal requirements. Australia and Singapore apply this approach. In Australia, monitoring is limited and guided by recommendations [57], while Singapore monitors treated effluent in the NEWater program, with results suggesting the absence of pharmaceutical substances in drinking water [58].

Level 3 is characterized by limited interventions, with occasional monitoring and minimal legal support. In the United States, the EPA uses the Unregulated Contaminant Monitoring Rule (UCMR) to collect data on unregulated contaminants, but UCMR4 does not include pharmaceuticals [59]. In the United Kingdom, businesses that generate pharmaceutical waste require permits from the Environment Agency [60], and in Denmark, hospitals need permits to discharge wastewater containing pharmaceuticals [54,61]. The Indian government plans to regulate antibiotic discharges [62].

On the other hand, pharmaceutical and personal care products have become a threat to aquatic environments, including bottled water, which has become a popular source of drinking water because of its taste and portability [63]. Studies on pharmaceutical contaminants in bottled water have revealed the presence of sulfonamides at concentrations of 9–80 ng/L [64], as well as other pharmaceutical substances such as metformin, salicylic acid, and gabapentin, detected only above the detection limit [65]. Salicylic acid, ketoprofen, and caffeine were identified in concentrations below 1 ng/L in natural mineral water samples [66]. Other analyses detected nicotine between 7–15 ng/L, but most substances analyzed were below the quantification limit [67]. A 2020 study [68] analyzed 92 antibiotics in 14 brands of bottled water, reporting average concentrations between 180 and 666 ng/L.

Table 1 presents the main classes of pharmaceutical products identified in environmental water samples detected using various electrochemical sensors and biosensors.

Table 1. Main classes of drugs and their representatives detected in water samples using different electrochemical sensors and biosensors.

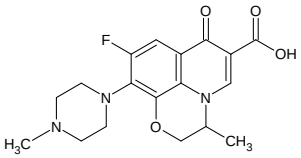
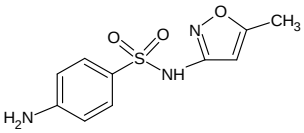
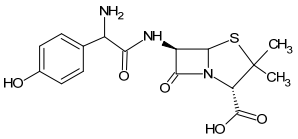
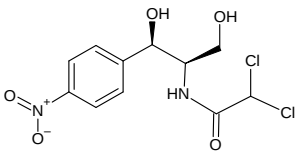
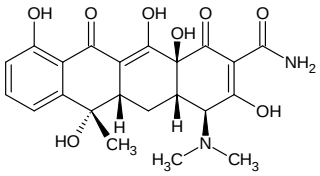
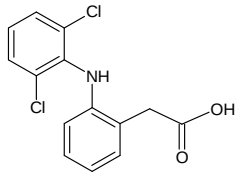
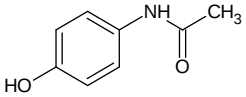
Pharmacological Class	Analyte	Electrochemical (bio)Sensor	Electrochemical Processes That Generate the Analytical Signal	Environment Samples	Ref.
Antibiotics	Ofloxacin 	CPE-based cyclodextrine and graphene oxide	Oxidation	River water	[69]
	Sulfametoxazol 	Ascorbic acid reduced graphene oxide-modified electrodes	Oxidation (diffusion-controlled process)	Lake water, tap water,	[70]
	Amoxicillin 	Impedimetric aptasensor-based TiO_2-g-C_3N_4 and gold nanoparticles	Oxidation	Wastewater	[71]
	Cloramphenicol 	Palladium nanoparticles decorated graphene oxide nanocomposite-modified glassy carbon electrode	Redox process (diffusion-controlled process)	Tap water	[72]
	Tetracicline 	Photoelectrochemical aptasensor for tetracycline-based on Au/BiOI composites	Photoinduction (separation of electron-hole pairs)	Water	[73]
Antiinflamatory, analgesic and antipyretic drug	Diclofenac  Paracetamol 	Poly(diallyldimethy lammonium chloride) functionalized graphene	Oxidation Quasi-reversible redox process	Lake water	[74]

Table 1. *Cont.*

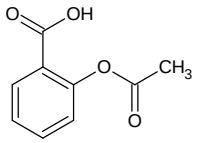
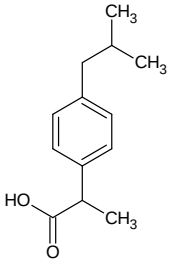
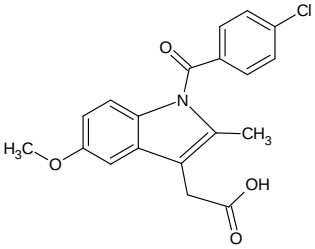
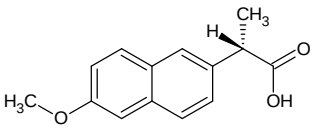
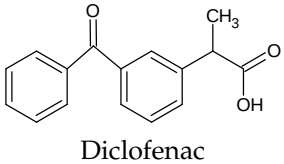
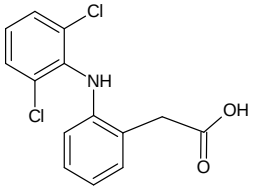
Pharmacological Class	Analyte	Electrochemical (bio)Sensor	Electrochemical Processes That Generate the Analytical Signal	Environment Samples	Ref.
Antiinflamatory, analgesic and antipyretic drug	Aspirin 				
	Ibuprofen 	Nitrogen-doped carbon nanodots and nanosized cobalt phthallocyanine conjugate-modified glassy carbon electrode	Oxidation	Wastewater	[75]
	Indomethacin 				
	Naproxen 	Photoelectrochemical Sensor based on a bismuth vanadate/copper oxide nanocomposite	Oxidation	Sewage water samples	[76]
	Ketoprofen 	Carbon paper electrode	Reversible redox process	Wastewater	[77]
	Diclofenac 	Enzyme-less electrochemical displacement heterogeneous immunosensor	Reversible redox process	Tap water	[78]

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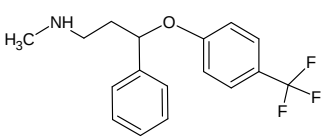
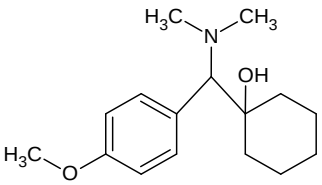
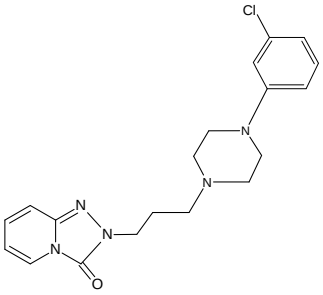
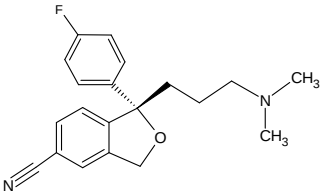
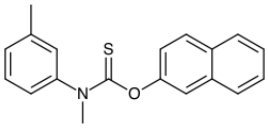
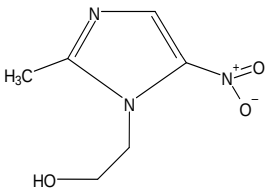
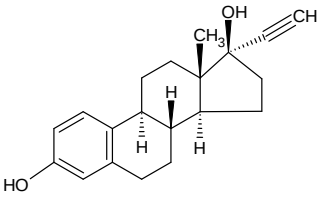
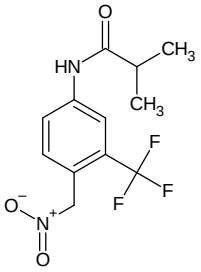
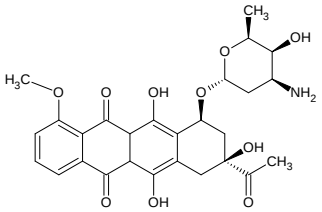
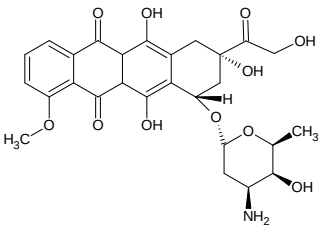
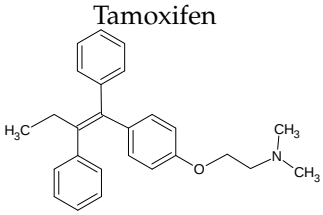
Pharmacological Class	Analyte	Electrochemical (bio)Sensor	Electrochemical Processes That Generate the Analytical Signal	Environment Samples	Ref.
Anti-depressants	Fluoxetine 	Carbon nanofiber-epoxy composite electrode	Oxidation (irreversible process)	Water	[79]
	Venlafaxine 	Screen-printed electrochemical sensor based on Fe_3O_4@cellulose nanocrystals/Cu nanocomposite	Oxidation (diffusion-controlled process)	Water	[80]
	Trazodone 	Molecularly Imprinted Polymer on a Screen-Printed Electrode	-	Tap water	[81]
	Citalopram 	Molecularly Imprinted Polymer on a Screen-Printed Electrode	-	Spiked river water	[82]
Anti-fungal	Tolnaftate 	Screen-printed carbon electrode	Oxidation (irreversible process)	Other samples	[83]
	Metronidazole 	Boron-doped diamond electrode pretreated	Oxidation (irreversible process) (diffusion-controlled process)	Other samples	[84]

Table 1. *Cont.*

Pharmacological Class	Analyte	Electrochemical (bio)Sensor	Electrochemical Processes That Generate the Analytical Signal	Environment Samples	Ref.
Hormones and nonsteroidal antiandrogen	17β-estradiol 	Electrochemical label-free aptasensor-based bi-functional graphene	-	Water	[85]
	Flutamide 	Exfoliated graphitic-carbon nitride-modified glassy carbon electrode	Reduction (adsorption controlled process)	Water samples	[86]
Anti-cancer drug	Daunorubicin 	Amino-terminated G4 PAMAM dendrimer-modified graphite sensors	Oxidation	-	[87]
	Doxorubicin 	Carbon black, copper nanoparticles, and Nafion-modified glassy carbon electrode	Oxidation (adsorption controlled process)	Water river samples	[88]
	Tamoxifen 	MIPs sensor based on an O-phenylenediamine–resorcinol mixture	Oxidation	-	[89]

2.2. Electrochemical Methods for Removal of Pharmaceutical Contaminants

The presence of pharmaceuticals in wastewater and surface waters poses significant concerns for environmental and public health. Traditional water treatment systems, such as common wastewater treatment plants or the use of chlorine dioxide, have limited efficiency in removing pharmaceutical compounds; therefore, advanced solutions are needed for the effective treatment of these pollutants. Electrochemical technologies, such as Advanced Electrochemical Oxidation Processes (EAOPs), offer promising solutions

for treating water contaminated with pharmaceutical substances. These processes utilize oxidation anodes to generate heterogeneous radicals, contributing to the decomposition of persistent compounds [90].

For example, direct oxidation, using Ti/PbO_2 as the anode and a carbon plate as the cathode, was the primary mechanism for the electrochemical removal of carbamazepine from pharmaceutical wastewater [91]. Another approach for the degradation of carbamazepine is indirect electro-oxidation with active chlorine, using Nb/BDD or Ti/IrO_2 as anodes. The oxidation process is more effective with Nb/BDD because of the increased production of •OH radicals and species HClO, Cl•, and ClO^- [92]. In another study, BDD electrodes were used for the electrochemical oxidation of pharmaceuticals such as iopromide, sulfamethoxazole, 17-alpha-ethinylestradiol, and diclofenac from simulated wastewater and hospital effluent, with slower degradation of the transformation products [93]. Chang et al. [94] utilized multi-walled carbon nanotubes, Pt nanoparticles, and PtRu alloys for synthesizing efficient anodes from commercial conductive glass for ibuprofen degradation. The addition of MWCNTs significantly reduced the grain size of the electrocatalyst, and the bimetallic PtRu anode demonstrated superior efficiency because of the high capacitance, catalytic activity at less positive voltages, and increased stability. Another example is the removal of contaminants from wastewater through the electro-Fenton process, which involves continuous H_2O_2 generation at a carbonaceous cathode, the addition of a Fe^{2+} catalyst, and the reduction of Fe^{3+} to Fe^{2+}. This process has proven effective in removing anti-inflammatory drugs (diclofenac, ibuprofen, and naproxen), achieving up to a 90% reduction in total organic carbon after 3 h of reaction [95].

Thus, electrochemical processes prove to be a viable solution for advanced treatment of water contaminated with pharmaceuticals, contributing to environmental protection and human health.

3. Electrochemical Sensors for Pharmaceutical Contaminant Detection

3.1. Operating Principles and Modifying Materials of Electrochemical Sensors

Electrochemical sensors are essential tools for detecting chemical compounds and operate based on the interaction between the analyte and the active element of the sensor, followed by the transduction, recording, and display of the signal. The electrochemical cell consists of a specific configuration made up of a properly modified working electrode, a reference electrode, and a counter electrode, all immersed in suitable electrolytes to facilitate electrochemical reactions at the surface of the working electrode [96].

Electrochemical sensors facilitate electrochemical reactions at the electrode interface, producing changes in current or potential due to redox reactions. They are classified into potentiometric and amperometric types, with the former measuring potential changes and the latter focusing on current variations [97]. Potentiometric sensors measure the potential difference between electrodes, and this value is proportional to the analyte concentration, while amperometric sensors measure the current generated by electrochemical oxidation or reduction reactions [97]. The current intensity measured is directly proportional to the analyte concentration in the solution. Impedimetric sensors measure changes in the electrical impedance (resistance) of the solution caused by the presence of the analyte. Impedance variations are correlated with the analyte concentration and can be used to determine its quantity in the solution [98]. These sensors are typically designed to provide high sensitivity and selectivity for detecting specific compounds, which is crucial for identifying pharmaceutical contaminants that may affect environmental quality and safety. Detection in electrochemistry is based on a series of fundamental processes that transform the presence of an analyte into a measurable electrical signal. The mechanism begins with the transport of the analyte to the surface of the electrode, achieved through diffusion,

migration, or convection, depending on experimental conditions and solution composition. Once the analyte reaches the electrode, it interacts with its surface [99]. These interactions can be physical, through weak forces, or chemical, through the formation of covalent or coordination bonds. Under the applied potential, the analyte undergoes an electrochemical reaction, usually oxidation or reduction. This transformation involves the transfer of electrons between the analyte and the electrode surface, generating an electric current. The amplitude of this current is proportional to the analyte concentration and reflects the sensor's ability to detect substances under varying conditions. The schematic of the detection mechanism of an electrochemical sensor is shown in Figure 2.

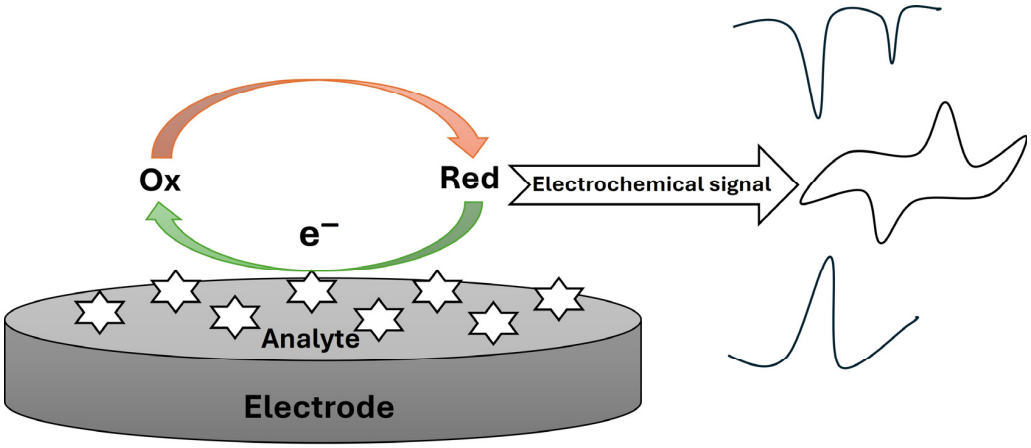


Figure 2. Schematic representation of the detection mechanism of an electrochemical sensor.

The performance of an electrochemical sensor significantly depends on the properties of its active surface [100]. Materials used for modification contribute to improving electron transfer, reducing the overpotential required for the reaction, and minimizing interferences [101]. These enhancements enable more precise and stable detection of the analyte in diverse environmental conditions. Therefore, the electrochemical detection mechanism represents a complex process involving a series of interdependent steps, each influencing the final performance of the sensor. The adaptation of materials and the optimization of experimental conditions play an important role in maximizing the sensitivity and reliability of the device. Advanced technologies, such as the use of nanomaterials and chemically modified electrodes, facilitate the electrochemical process and enhance sensor performance, enabling the detection of contaminants at extremely low concentrations and within a very short time [102]. For example, the use of nano-metallic and nano-magnetic structures mixed with polymers on screen-printed electrodes (SPE) allows for the customization of active surfaces for environmental and clinical sample analysis [103]. These materials improve the sensitivity and selectivity of sensors, facilitating the detection of emerging contaminants that are not typically monitored in water treatment stations but may negatively impact the environment and human health [104]. Furthermore, screen printing methods simplify the mass production of sensors, making them stable and affordable [105].

3.2. Performance of Electrochemical Sensors for Pharmaceutical Contaminant Detection in Water Samples

3.2.1. Electrochemical Sensors Based on Carbon Nanomaterials

Nanomaterials have revolutionized electrochemical detection platforms, especially for pharmaceutical compounds. Carbon-based materials have become popular due to their high conductivity and ability to enhance electron transfer and adsorption of organic molecules [106]. A notable example of an electrochemical sensor for the simultaneous detection of two pharmaceutical substances is the carbon black paste electrode (CBPE)

modified with hierarchical porous carbon and α-cyclodextrin, used by Wong et al. [107]. This synergistic combination significantly improved the electrochemical properties of the sensor, allowing for the detection of dipyrone and codeine at nanomolar levels. The method is distinguished by easy surface preparation, high sensitivity, rapid response, and the ability to refurbish the surface by polishing. Successfully applied to river water samples, the sensor provided recovery rates of about 100%, demonstrating remarkable potential for the simultaneous detection of these analytes. Another innovative electrochemical sensor was developed by Henrique et al. [108] for the detection of sulfamethoxazole, using a conductive ink based on graphite and shellac applied to an impermeable paper substrate. This sensor has a wide linearity range, competitive detection limits, and recoveries between 91% and 110%, with no significant interferences from the sample matrix. A very recent approach for detecting and combating antibiotic contaminants in water involves a new ultrasensitive electrochemical sensor obtained by integrating coral-type nanostructures made of gold-silver alloy with a CPE functionalized with multi-wall carbon nanotubes and choline chloride nanocomposites. This sensor was designed for the simultaneous detection of rifampicin and norfloxacin residues in water samples, with extremely low detection limits [109]. The excellent linearity of the sensor over a wide range of concentrations can be seen in Figure 3.

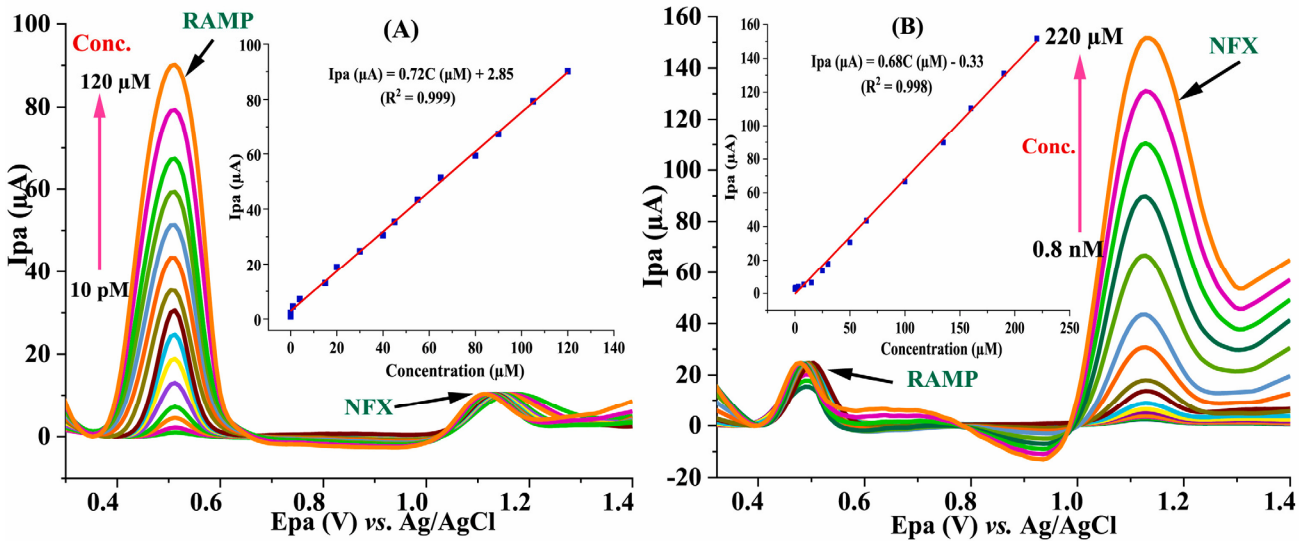


Figure 3. SWV recordings for different concentrations of rifampicin (**A**) and norfloxacin (**B**) (Reprinted with the permission of the publisher [109]).

Diclofenac, a widely used nonsteroidal anti-inflammatory drug, has attracted the attention of researchers because of its potential contamination of water sources. Therefore, a screen-printed carbon electrode modified with multi-walled carbon nanotubes functionalized with carboxyl groups (SPCE/MWCNTs-COOH) was developed for the determination of this pharmaceutical in water samples, applying differential pulse voltammetry via adsorptive stripping. The high performance of the modified sensor was demonstrated by excellent responses and a linear range between 0.1 and 10.0 nmol L^{-1}. Without the need for sample pre-treatment, the analysis of Vistula river water showed the presence of diclofenac at a concentration of 0.42 ± 0.08 nmol L^{-1} in a sample collected from a sewage discharge point. The intra-day and inter-day precision of the SPCE/MWCNTs-COOH sensor were 0.7% and 2.1%, respectively, and the reproducibility, assessed by recording DPAdSV curves with three commercially available electrodes, showed an RSD of 2.9%, confirming the satisfactory repeatability and acceptability of the sensor [110].

3.2.2. Sensors Modified with Carbon Nanomaterials, Metallic Nanoparticles and Metal Oxides

Similar to other carbon-based nanomaterials, graphene and graphene oxide offer advantages for electrochemical detection through doping with heteroatoms, which can induce surface charges useful for electrostatic interactions [111]. Metal nanoparticles, due to their catalytic properties, are often used in pharmaceutical detection, with controlled sizes and shapes for catalytic synergy [112,113]. For example, the simultaneous detection of amitriptyline and carbamazepine was achieved using a screen-printed carbon electrode modified with a $MoSe_2$/graphene oxide reduced (rGO) nanocomposite. The $MoSe_2$/rGO nanocomposite was fabricated by dissolving graphene oxide powder in deionized water and ethanol, followed by a chemical reaction in an autoclave to obtain the desired precipitate. The $MoSe_2$-rGO/SPCE sensor demonstrated remarkable electrochemical performance with high sensitivity and an effective detection limit (0.007 µM). Several samples, including tap water, were tested using the standard addition method, achieving recovery rates between 96.0% and 104.3% [114].

The CuxrGO_{1-x}/GCE sensor, developed by electroplating copper nanoparticles onto rGO for metformin quantification, also shows great potential for surface water analysis. The sensor demonstrated uniform dispersion of Cu nanoparticles in the rGO matrix, leading to a significant increase in anodic current and a reduction in the oxidation overpotential of the antidiabetic drug. Electrochemical analysis revealed an average particle size of 70 nm and an electrochemically active surface area of 2 cm^2 for the CuxrGO_{1-x}/GCE electrodes. The Cu(II)/Cu(I) redox couple mediated the oxidation of the imino group in metformin, achieving a maximum peak current sensitivity in the linear range of 2–70 µM, with a detection limit in the micromolar range. The sensor also demonstrated high reproducibility, selectivity, and analytical recovery, making it a suitable alternative for the electrochemical detection of metformin using a non-precious metal-based catalyst [115].

The improper use of ciprofloxacin, a synthetic antibacterial agent, can contaminate the environment and increase the risks of antibiotic resistance, making its identification and quantification necessary. Zokhtareh et al. [116] developed an efficient electrochemical sensor for the detection of ciprofloxacin using a carbon paste electrode modified with graphene nanosheets and eco-friendly Fe_3O_4 nanoparticles. Figure 4 illustrates the electrochemical reaction mechanism of ciprofloxacin at the GR/Fe_3O_4NPs electrodes, highlighting the interactions between the cationic form of the fluoroquinolone and the negative groups on the electrodes, which facilitate its oxidation through irreversible reactions involving two electrons and two protons.

The modification of the electrodes significantly increased the electrocatalytic activity and sensitivity, achieving a detection limit of 1.8 nM. The robustness of the sensor was demonstrated by its stable performance across a range of pH values (4 to 8) and temperatures (20 °C to 40 °C), with a relative standard deviation (RSD) of less than 5% under these varying conditions. It also demonstrated excellent repeatability with an RSD of 1.6% after five consecutive DPV analyses using the same electrode and good reproducibility with an RSD of 4.31% after measurements with five separate electrodes [116]. By using eco-friendly materials, this sensor not only enhances the detection of ciprofloxacin but also promotes sustainable environmental monitoring solutions.

In the same year, Mariappan et al. [117] demonstrated the good efficiency of a nanomaterial composed of zinc oxide and tungsten ($ZnWO_4$), prepared through a simple hydrothermal synthesis, to create a composite with black carbon for the determination of the same pharmaceutical compound. The crystalline structure and form of the materials were confirmed by advanced spectroscopic characterizations such as XRD, FTIR, FESEM, and elemental mapping EDX. The GCE/$ZnWO_4$/CB electrode shows satisfactory sensitivity and

detection limits, thus demonstrating an efficient electrocatalytic reaction in ciprofloxacin detection. In comparison with the previous study using graphene nanomaterials and iron oxide nanoparticles, the GCE/$ZnWO_4$/CB electrode stands out with a more accessible construction method and lower cost, although interference studies were not conducted, and the recoveries for real samples were somewhat lower. Both studies highlight the importance of pH in the ciprofloxacin detection process.

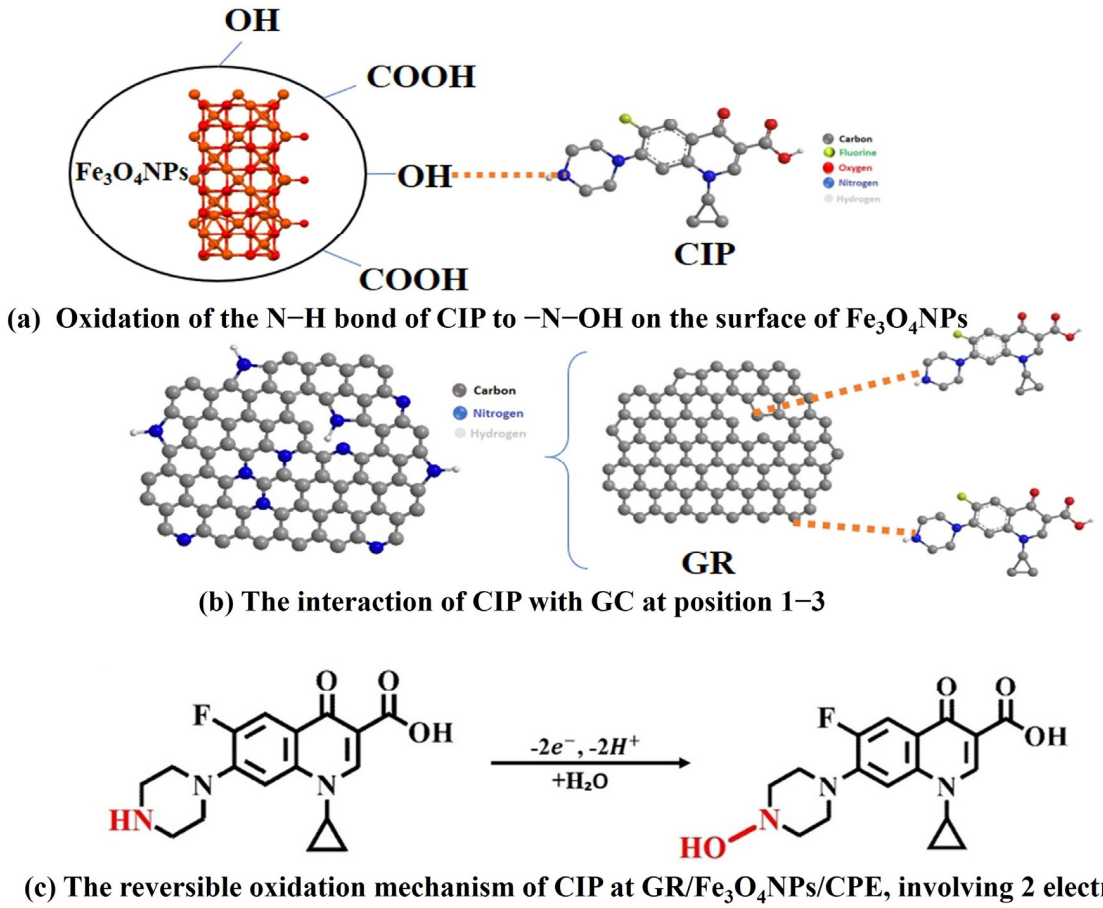


Figure 4. A possible mechanism of ciprofloxacin oxidation using GR/Fe_3O_4NPs/CPE (Adapted with the permission of the publisher [116]).

In another study, the simultaneous determination of ciprofloxacin and paracetamol in river water samples was pursued. The chosen graphite electrode was cleaned and modified with a TiO_2 sol, mesoporous carbon CMK-3, and gold nanoparticles. The TiO_2 sol was obtained by acid hydrolysis; gold nanoparticles were synthesized by reducing tetrachloroic acid, and mesoporous carbon CMK-3 was prepared by hard templating. The final composite was applied to the electrode and deposited above a saturated solution of disodium phosphate diluted with water (1:1, v/v) and then in a phosphate buffer solution to allow the polymerization of the TiO_2 sol. This configuration enhances the conductivity and stability of the sensor, providing high precision and accuracy in the analysis of samples from the Wisła and Racławka rivers, as well as from wastewater, demonstrating high precision and accuracy with minimal relative errors. The sensor demonstrated good reproducibility with RSD values of 2.1% for ciprofloxacin and 4.5% for paracetamol, indicating stable performance across five independently prepared sensors, and its robustness was confirmed by consistent analytical parameters over two weeks of use [118].

The frequent use of amoxicillin has drawn the attention of researchers in Brazil. They developed a sensor by modifying a glassy carbon electrode with reduced graphene and

Nafion for the determination of amoxicillin in river water samples. The electrode modifications improve conductivity and selectivity, facilitating the catalysis of the electrochemical reaction. The analysis method used is square wave voltammetry, which allows the detection of this antibiotic at low concentrations, with detection limits of 0.36 mmol L^{-1}, exhibiting good selectivity in the presence of benzylpenicillin. The sensor was validated using rigorous statistical parameters, demonstrating good linearity, precision, and accuracy comparable to chromatographic methods [119].

3.2.3. Sensors Modified with Conducting Polymers

Furthermore, conductive polymers, integrated into synthetic composites along with carbon nanomaterials, offer high conductivity, increase sensor selectivity, and may facilitate the adsorption of organic molecules [120]. For example, PEDOT: PSS (Poly(3,4-ethylenedioxythiophene)-poly(styrenesulfonate)) can reduce graphite layer separation, thus improving the electrical conductivity and stability of electrochemical sensors. Moreover, studies show that nanoparticles of noble metals could bring significant benefits to flexible graphene sheets by increasing their resistance [121,122]. In 2022, an electrochemical sensor modified with a uniform hybrid film of reduced graphene oxide (rGO), PEDOT, and gold nanoparticles was developed for the detection of acetaminophen, a common pharmaceutical pollutant in aquatic environments. The hybrid film was created by electrochemical reduction of a suspension of rGO and PEDOT, resulting in a compact and adherent layer on the GCE electrode surface. Subsequently, gold nanoparticles were electrochemically deposited, increasing the active surface for acetaminophen detection. The sensor exhibited enhanced sensitivity, with a detection limit of 532 nM, and greatly improved the electronic transfer rate. A slight decrease in response (2–4%) was observed when interferents such as dopamine, citric acid, or caffeine were added to the solution [123].

In another study, the PANI-AgBr/SPE sensor was modified to enhance the detection of amoxicillin in drinking water and lakes using cyclic voltammetry and chronoamperometry. This modification involved the integration of AgBr particles with polyaniline (PANI) through an electrochemical deposition process, which led to increased conductivity due to ionic interactions, generating a favorable environment for redox reactions. The process of obtaining PANI-AgBr can be seen in Figure 5. Experiments showed high selectivity towards the analyte, with negligible differences in the presence of interferents such as organic acids and inorganic compounds. The sensor's stability and repeatability were evaluated, demonstrating consistent performance over 30 days, highlighting its ability to monitor amoxicillin in real-time across various water samples. The PANI-AgBr sensor demonstrated excellent robustness, maintaining stable amperometric responses and selectivity for amoxicillin even after 30 days of use, with minimal decay (~30%) [124].

Another modifier polymer is poly(alizarin red S), which was used to construct an amperometric sensor for the determination of hydrazine in various water samples. The modification of the graphite electrode was achieved by recording several voltammetric cycles in a solution of alizarin red and $HClO_4$, thus optimizing the sensor's performance for hydrazine detection. Electropolymerization led to a reduction in the oxidation potential and an increase in the oxidation current intensity of hydrazine, improving the sensor's sensitivity and selectivity. After optimizing parameters such as monomer concentration and the number of electropolymerization cycles, the sensor was tested for stability and interference, showing excellent performance in hydrazine detection in real water samples. The recovery values obtained for hydrazine in various water sources ranged from 95% for drinking water to 108% for seawater, demonstrating the efficiency of the modified sensor in the amperometric determination of hydrazine in complex samples [125].

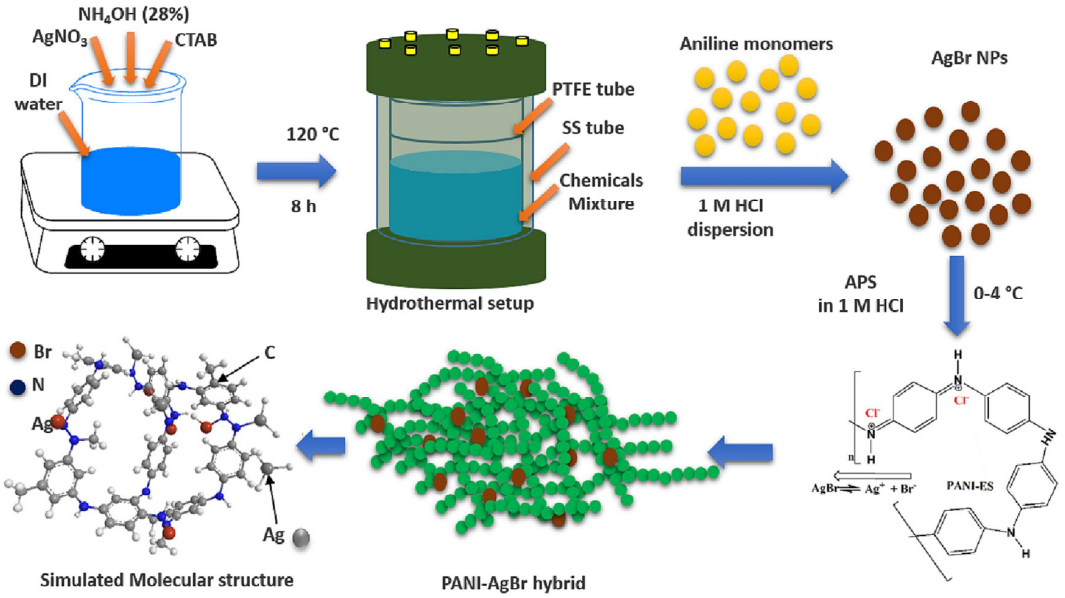


Figure 5. Stages of nanomaterial synthesis for the construction of PANI-AgBr/SPE (Reprinted with the permission of the publisher [124]).

3.2.4. Sensors Based on Molecularly Imprinted Polymers

Another sensor construction technique is the modification with molecularly imprinted polymers (MIPs), which allow for the rapid and precise detection of the desired analyte by creating specific cavities formed according to the target molecule. A notable example of applying these polymers in diclofenac detection is an electrochemical sensor based on the porous structure of PEDOT generated by combining electrochemical polymerization with nanosphere-based lithography. This sensor was constructed on a GCE modified with polystyrene spheres and then coated with PEDOT in the presence of diclofenac, which served as the template molecule. After extracting the diclofenac and polystyrene spheres, the "honeycomb" type porous structure allowed increased accessibility of diclofenac molecules to the active surface. Electrochemical tests demonstrated a detection limit of 77.5 nM for diclofenac. The results suggest new research directions for electrochemical detection of diclofenac in water samples [126].

Simultaneous detection of diclofenac and codeine in water samples was achieved using a biomimetic sensor. The sensor was developed by depositing a layer of polyvinyl chloride (PVC) carboxylate onto a gold electrode. After activating the -COOH groups with EDC and NHS, a solution containing both pharmaceutical substances was deposited. Silver nanoparticles and methacrylic acid were then immobilized. Finally, an extraction step was performed to remove the target molecules. The sensor was tested using electrochemical techniques such as differential pulse voltammetry and electrochemical impedance spectroscopy, achieving a detection limit of 0.01 μg/mL for diclofenac and 0.16 μg/mL for codeine; however, simultaneous detection was affected by cross-reactivity. Using an electronic tongue to collect additional data on the samples and principal component analysis, promising results were obtained for classifying water samples containing both substances [127].

Another approach involved modifying a GCE through the electropolymerization of a copolymer of polyaniline (PANI) and o-phenylenediamine (o-PDA) aimed at ciprofloxacin detection. This modification strategy combines the advantages of a molecularly imprinted polymer with those of reduced graphene oxide (rGO), which enhances carrier mobility and contributes to the amplification of the electrochemical signal. The sensor displayed excellent electrochemical characteristics, highlighting an anodic and cathodic current that increased linearly with the scan rate, suggesting a diffusion-controlled electrochemical reac-

tion. Regarding analytical performance, the MIP/rGO/GCE exhibited a detection limit of 5.28×10^{-11} mol L^{-1} and a wide dynamic measurement range, demonstrating remarkable sensitivity and selectivity for ciprofloxacin in the presence of other antibiotics. Recoveries in real water samples ranged from 84.2% to 124.8%, highlighting the sensor's ability to detect ciprofloxacin in varied environmental conditions [128]. These results suggest significant potential for using the sensor in monitoring antibiotics in environmental samples.

3.2.5. Sensors Based on Covalent Organic Frameworks

Recently, the use of covalent organic frameworks (COFs) has been explored as a promising alternative for developing electrochemical sensors, especially in combination with conductive polymers. This combination not only enhances electrical conductivity but also provides a greater number of active sites, facilitating sensitive analyte detection. In a recent study, the TFAB-COF@PANI composite was developed as a sensitive electrochemical sensor for sulfamethoxazole detection using a simple in situ synthesis. The sensor was constructed by modifying glassy carbon electrodes with TFAB-COF@PANI, obtained by the in situ growth of TFAB-COF (a covalent organic framework formed by Schiff base condensation between 2,4,6-tris-(4-formylphenoxy)-1,3,5-triazine and 4-aminobenzohydrazide) on polyaniline (PANI), enhancing the electrochemical signal. The sensor was tested under optimal conditions at 50 μM sulfamethoxazole, with reproducibility evaluated over 10 successive measurements and long-term stability assessed by storing the sensor at 4 °C. Interference was tested with inorganic species (e.g., $CaCl_2$, KCl, K_2CO_3, and $NaNO_3$) at 100 times the sulfamethoxazole concentration and organic molecules (amoxicillin, ibuprofen, glutamate, chloramphenicol, urea, and glucose) at 50 times the sulfamethoxazole concentration, with no significant effect on analyte determination, yielding RSDs of less than 5%. The sensor demonstrated a wide linear range and low detection limit, being effective in analyzing water samples (Yellow River water and tap water). The sensor showed good repeatability, reproducibility, and stability, with sulfamethoxazole recoveries ranging from 99.84% to 103.53% [129].

3.2.6. Sensors Based on Metal–Organic Frameworks

Metal–organic frameworks (MOFs), similar to COFs, have diverse applications in gas capture, catalysis, and electrochemical sensors because of their porosity and adjustable chemical properties [130,131]. In this context, ZIF-8 (zeolitic imidazolate framework-8), a type of MOF, was used to construct an electrochemical sensor for the detection of sulfamethoxazole, bisphenol A, and sodium diclofenac. The sensor was constructed by depositing a layer of ZIF-8 onto a vitreous electrode. The ZIF-8 synthesis process involved forming a mixture of zinc, 2-methylimidazole, and a methanol solution, followed by centrifugation and drying. The ZIF-8 electrode was obtained by mixing ZIF-8 with isopropyl alcohol and Nafion solution, ensuring uniform dispersion on the electrode surface. Electrochemical evaluations were performed using differential pulse voltammetry and electrochemical impedance spectroscopy. The results showed that the ZIF-8 sensor had high sensitivity and a low detection limit for sulfamethoxazole, with recovery efficiency ranging from 95.3% to 103.0% in lake water samples and from 94.3% to 96.0% in tap water samples [132]. These performances suggest that ZIF-8 could be a promising tool for monitoring organic pollutants in real-world environments.

On the other hand, Shahsavari et al. [133] used a carbon paste electrode as a support for the Fe_3O_4/ZIF-67 nanocomposite and the ionic liquid 1-Butyl-3-methylimidazolium hexafluorophosphate, obtaining Fe_3O_4/ZIF-67/ILCPE for sulfamethoxazole detection. The presence of Fe_3O_4 magnetic nanoparticles enhances catalytic efficiency and solution stability. This sensor exhibits high electrocatalytic activity, selectivity, and sensitivity in real sample

analysis, including tap water, river water, and urine. The reproducibility of the constructed sensor was assessed through successive measurements, while its long-term stability was evaluated by storing it at 4 °C, yielding consistent results with an RSD of less than 5%. Table 2 summarizes the performance of electrochemical sensors modified with various nanomaterials for the detection of pharmaceutical substances in different water samples. These sensors were developed using advanced materials to enhance sensitivity, specificity, and overall detection performance. The information includes key parameters such as electrochemical methods used, detection limits (LOD), and the linear range, providing an overview of their efficiency in monitoring pharmaceutical contaminants in the environment.

Table 2. Performance of electrochemical sensors for pharmaceutical contaminant detection in water, including sensor type, materials used, detected substances, detection limits, and applications in water samples.

Sensor/Modifying Material	Method	Analyte Detected	LOD	Linear Range	Slope/Sensitivity	Type of Water Sample	Ref.
Carbon nanomaterials							
CBPE electrode modified with α-cyclodextrin and hierarchical porous carbon	SWV	Dipyrone Codeine	0.009 μmol L^{-1} 0.0095 μmol L^{-1}	0.46–35 μmol L^{-1} 0.5–38 μmol L^{-1}	0.04 μmol L^{-1} 0.049 μmol L^{-1}	River water (Tietê)	[107]
Electrochemical sensor based on shellac and graphite	DPV	Sulfamethoxazole	0.4 μmol L^{-1}	5.0 μmol L^{-1}–100 μmol L^{-1}	0.09 μA μmol L^{-1}	Well water Tap water	[108]
SPE modified with carboxyl functionalized multiwalled carbon nanotubes	DPAdSV	Diclofenac	0.028 nmol L^{-1}	0.1–10 nmol L^{-1}	0.040 A/nmol L^{-1}	Vistula river water	[110]
Carbon nanomaterials, metallic nanoparticles, and metal oxides							
Gold-silver alloy nanocoral integrated with CPE	SWV	Rifampicin Norfloxacin	2.7 pM 0.14 nM	14 pM–115 μM 0.9 nM–200 μM	0.73 μA/μM 0.68 μA/μM	River water Hospital Wastewater Tap water	[109]
SPE modified with $MoSe_2$ and reduced graphene oxide	DPV	Amitriptyline Carbamazepine	0.007 μM	0.02–380.0 μM	0.1283 μA/μM	Tap water Other samples	[114]
Copper nanoparticle-modified reduced graphene oxide electrode	DPV	Metformin	0.09 μM	2–70 μM	1.375 μA/μM	Tap water River water, Wastewater	[115]
Graphene and Fe_3O_4 nanostructure-modified CPE	DPV	Ciprofloxacin	1.8 nM	0.01–20 μM	47.91 μA $μM^{-1}$ cm^{-2}	Antibiotic plant effluent River water (Neka Rood)	[116]
GCE modified with $ZnWO_4$/CB	DPV	Ciprofloxacin	0.02 μM	0.02–120 μM	1.71 μA $μM^{-1}$ cm^{-2}	Tap water River water Tap water	[117]
graphite electrode modified with TiO_2, CMK-3, and gold nanoparticles	CV	Ciprofloxacin Paracetamol	0.108 μM 0.210 μM	1–10 μM and 10–52 μM	15.93 μA $μM^{-1}$ 11.56 μA $μM^{-1}$	Wisła river Racławka river CRM of wastewater	[118]
Graphene and ZnO Nanorod-modified GCE	DPV	Sulfamethoxazole Trimethoprim	0.4 μmol/L 0.3 μmol/L	1–220 μmol/L 1–180 μmol/L	0.0987 μA/μmol/L 0.0269 μA/μmol/L 0.4123 μA/μmol/L 0.0746 μA/μmol/L	Tap water Lake water	[134]
A noble metal doped onto an oxidized one-dimensional (1D) single walled carbon nanotube and abundant oxygen functional groups presenting two-dimensional (2D) gra phene oxide	CV	Naproxen	14 nM	100 nM–15.6 mM	1.7484 μA $μM^{-1}$ cm^2	Tap water Pond water Sea water River water	[135]
Conductive polymers							
GCE modified with reduced graphene oxide and Nafion	SWV	Amoxicillin	0.36 mmol L^{-1}	1.8–5.4 μmol L^{-1}	3.5 μA $μmol^{-1}$ L	River water (Guarapuava)	[119]
Hybrid film of rGO, PEDOT, and gold nanoparticles	-	Acetaminophen	532 nM	1–9 μM	-	-	[123]
polyaniline-AgBr	CV	Amoxicillin	0.728 nM 0.193 nM	0.193–0.855 nM	40 μA/nM/cm^2	Tap water Lake water	[124]
Poly(alizarin red S)-modified pencil graphite electrode	CV	hydrazine	0.28 μM	1.0–600 μM	833 μA mM^{-1} cm^{-2}	Drink water Tap water Sea water	[125]

Table 2. *Cont.*

Sensor/Modifying Material	Method	Analyte Detected	LOD	Linear Range	Slope/Sensitivity	Type of Water Sample	Ref.
			MIP				
Molecular imprinting in conductive polymer film on GCE	DPV	diclofenac	77.5 nM	0.1–0.5 μM	0.4285 μA/μM	Water	[126]
(biomimetic-MIP sensor)		Diclofenac Codeine	0.01 μg/mL 0.16 μg/mL	0.001–500 μg/mL	-	Water samples simultaneously containing both these drugs	[127]
Molecularly Imprinted Copolymer of Polyaniline–Poly(o-phenylenediamine) coated on a rGO-modified GCE	DPV	Ciprofloxacin	0.00005 μmol L^{-1}	0.001–0.5 μmol L^{-1}	-	Tap water Pond water	[128]
CPE modified with magnetic MIP	SWV	Amoxicillin	0.75 μmol L^{-1}	2.5–57 μmol L^{-1}	0.24 μA μmol^{-1} L	River water	[136]
			COF				
Electrochemical sensor based on COF materials on polyaniline	DPV	Sulfamethoxazole	0.107 μM	1–450 μM	-	Tap water Yellow river water	[129]
			ZIF				
Zeolitic imidazolate framework-8 (ZIF-8) electrode	DPV	Sulfamethoxazole	25.7 nM	0.01–40.0 μM	0.05225 μA/μM	Lake water Tap water	[132]
CPE modified with Ionic Liquid and Fe_3O_4/ZIF-67	-	Sulfamethoxazole	5.0 nM	0.01−520.0 μM	12.718 $\mu A \cdot s^{-1/2} \cdot mM^{-1}$	Tap water River water	[133]

3.3. Advantages and Limitations of Electrochemical Sensors

Electrochemical sensors offer multiple advantages that make them extremely useful for detecting pharmaceutical substances in complex environments, such as surface waters. Carbon-based nanomaterials, such as multi-walled carbon nanotubes, enhance stability and electron transfer rate, thereby contributing to a more stable and reproducible signal over the long term. This is a crucial feature for sensors used in variable environments, such as wastewater, where conditions can differ significantly. On the other hand, bimetallic alloys, for example, provide superior electrical conductivity and improve catalytic activity, leading to greater sensor sensitivity and selectivity. These alloys can be highly beneficial for applications requiring the detection of very low concentrations of emerging pollutants in water, where performance enhancement is essential.

Moreover, functionalizing sensors with carboxyl, amino, or thiol groups significantly improves the electron transfer process and increases the electrode's active surface area, facilitating more efficient detection. Most of these sensors can accurately detect pharmaceutical substances in water samples, demonstrating much better performance in terms of reproducibility and measurement accuracy compared with non-functionalized sensors. Screen-printed electrodes, compared with carbon paste electrodes, allow for rapid, cost-effective production, making them ideal for environmental analysis applications. The combination of reduced graphene oxide and Nafion is an example of an advanced material used for building sensors to detect antibacterial substances such as amoxicillin. Reduced graphene oxide, with its high electron mobility, combined with Nafion, helps increase the sensor's sensitivity and selectivity. This combination enhances the catalytic reaction and allows for rapid and precise detection of pharmaceuticals in water samples, presenting an important opportunity for analyzing pharmaceutical substances in the environment.

However, electrochemical sensors are not without limitations. Interference from complex environments, such as the presence of ions or other contaminants in water samples, can affect the electrochemical signal. Additionally, environmental conditions, such as temperature or pH, can influence sensor performance, making them sensitive to changes in the environment in which they are used. This vulnerability to external conditions can lead to reduced measurement accuracy and may require periodic adjustments to maintain precision. Furthermore, the long-term stability of sensors can be an issue. Certain modify-

ing nanomaterials may degrade or become contaminated over time, necessitating frequent recalibration of the sensors to ensure optimal performance. Additionally, the functionalization or synthesis of nanocomposites may complicate the manufacturing process and increase costs, even though the results provided are remarkable. Therefore, electrochemical sensors offer a wide range of advantages, including high sensitivity, enhanced stability, and excellent performance in complex environments, but they also have limitations, such as sensitivity to interference and the need for regular maintenance. Adapting sensors for various applications and environmental conditions continues to be a significant challenge in research and development.

4. Electrochemical Biosensors for the Detection of Pharmaceutical Contaminants

4.1. Operating Principles

Considered a receptor–transducer device, the term "biosensor" dates back to the 1960s with the works of Clark and Lyons, proving capable of providing a series of quantitative analytical data through the use of a biological recognition element [137,138]. The widespread success of biosensors began with the amperometric detection of glucose for diabetes monitoring [139]. Biosensors were then developed for detecting other chronic conditions such as cancer [140], viruses, for example, Influenza viruses [141], Hepatitis virus [142], Severe Acute Respiratory Syndrome Coronavirus [143], or bacteria. Numerous biosensors have been applied to food samples or pharmaceutical products for quantifying a target analyte [144,145]. Recently, due to global concerns about environmental pollution, many biosensors have been developed for detecting pesticides, heavy metals, and pharmaceutical contaminants in environmental samples to monitor environmental quality [146,147].

The development of a biosensor involves the integration of several components: a bioreceptor (enzyme, cell, antibody, tissue, microorganism, aptamer, or nucleic acid), a transducer (nanomaterial or semiconductor material), and an electronic system that includes an amplifier, signal processor, and interpretation software, all designed to detect the molecule of interest [148]. For example, Figure 6 shows the components of an electrochemical biosensor and its operating principle.

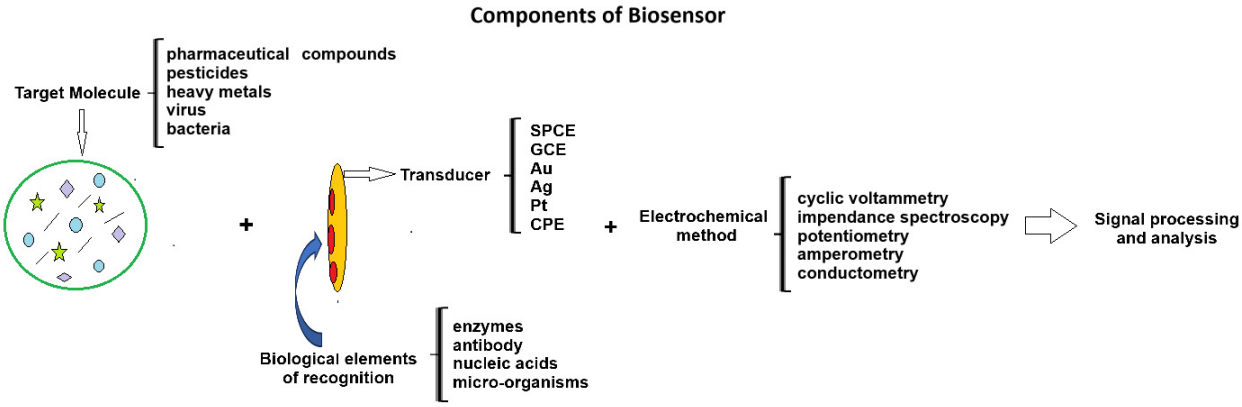


Figure 6. Structure of an electrochemical biosensor (Adapted with the permission of the publisher [148]).

These versatile devices primarily aim to detect changes in interactions occurring at the electrode surface, where the biological recognition factor interacts selectively with the target analyte, thereby generating an electric signal [149]. This signal is transmitted to the processor via the transducer.

While conventional electrochemical sensors rely on chemical or physical changes (e.g., redox reactions), biosensors integrate bio-specific recognition elements such as enzymes, antibodies, or DNA sequences to enhance selectivity. These elements enable the sensor to identify specific targets, even in complex water samples, mimicking natural biological processes. For instance, enzyme-based biosensors are widely used for detecting pharmaceutical contaminants such as antibiotics. The enzyme catalyzes a reaction with the analyte, generating a measurable electrochemical signal. Transduction can be direct (where the enzyme transfers electrons to electrodes) or indirect (where mediators facilitate electron transfer). Mediators, such as ferrocene, are small and mobile, allowing for rapid reactions. Carbon nanotubes can replace mediators, guiding electrons between the enzyme and electrodes [149]. Similarly, antibody-based biosensors operate based on the antigen-antibody affinity mechanism, where antibodies specifically recognize and bind to antigens present in samples, enabling precise and sensitive detection and quantification [150]. When a DNA probe (a biorecognition element) is immobilized on the electrochemical transducer, DNA-based biosensors are created. These are used for detecting molecules that bind to immobilized DNA through interactions that induce changes in the structure and electrochemical properties of DNA, which are then transformed into an electric signal. DNA-based biosensors can recognize genetic markers of contaminants, expanding their applications in environmental monitoring [151].

Unlike conventional sensors, biosensors can distinguish specific analytes within mixtures, making them more suitable for the complex matrix of bottled water. This enhanced capability is essential for addressing pharmaceutical contaminants present at trace levels in water [68]. The integration of bio-specific recognition elements highlights a significant advancement in electrochemical sensor technology, enabling precise and selective detection in complex environments.

The most versatile biosensors are electrochemical ones, owing to the advantages they offer, which are described in the following section.

4.2. Types of Electrochemical Biosensors and Biological Materials Used

The specialized literature mentions many classification criteria for electrochemical biosensors, all based on the type of bioreceptor and the type of transducer [152]. Thus, they are distinguished according to the type of transducer into amperometric, potentiometric, voltammetric, conductometric, and impedance-based electrochemical biosensors [153]; moreover, depending on the type of bioreceptor, biosensors based on enzymes, antibodies, aptamers, cells, tissues, and nanobiosensors have been identified [154]. Their performance has evolved because of the use of nanomaterials with excellent properties (unique structure, chemical stability, high terminal conductivity, good reproducibility, and remarkable electrocatalytic activity), such as molecularly imprinted polymers, conducting polymers, metal nanoparticles, and multi-walled carbon nanotubes [155]. Equally important is the correct selection of the immobilization technique for the biological recognition element on the surface of the electrodes [156]. This aspect gives the biosensor specificity, where biomolecules must be able to retain their biological activity, function, and structure, as there is a risk of them becoming inactivated or removed if the technique used is not correctly selected. There are reversible (physical) and irreversible (chemical) immobilization methods [157]. Reversible methods include physical adsorption, electropolymerization, and microencapsulation, which mediate the attachment of enzymes to the electrode surface [158]. Direct covalent binding and covalent cross-linking are irreversible methods that play an important role in the formation of strong bonds. All of these nanomaterials mentioned are found in the construction of many biosensors used for detecting pharmaceutical compounds in environmental samples [159]. In the next section, these biosensors are classified based on the material used for modification.

4.3. Performance of Electrochemical Biosensors for the Detection of Pharmaceutical Contaminants in Water Samples

Numerous studies have been identified for the detection and quantification of pharmaceutical compounds using electrochemical biosensors, with a focus on water quality monitoring, which has significantly contributed to the development of various management strategies for environmental monitoring. Maintaining the same classification criterion as for sensors, namely according to the type of nanomaterial used, several biosensors are exemplified in Table 3, developed for the identification of pharmaceutical analytes.

4.3.1. Biosensors Based on Carbon Nanomaterials

A biosensor with excellent properties was proposed by Canevari et al. [160], where the target molecule was the hormone 17α-ethynylestradiol. The electrode, fabricated and modified with a hybrid material made of single-walled carbon nanotubes and nanocrystalline carbon quantum dots (SWCNT/C-dot), on which the enzyme laccase was immobilized, was analyzed through various methods, including electrochemical techniques such as DPV and SWV. An improvement in electron transfer of copper ions from the enzyme's active sites was demonstrated for the GCE/SWCNT/C-dot/Lac biosensor, along with excellent reproducibility and stability after ten successive measurements and five different electrodes. The detection limit obtained for this biosensor was 4.0 nmol L^{-1}, much better compared with the GCE/SWCNT/C-dot sensor. Testing on water samples was carried out without any prior treatment. All these results led the authors to conclude that the synergy achieved by combining SWCNT, C-dot, and the laccase enzyme increased the value of the electrochemical device through high sensitivity [160].

4.3.2. Biosensors Modified with Metallic Nanoparticles

A biosensor model useful for identifying pharmaceutical contaminants in water was developed by Okoth et al. [161]. The authors used graphene doped with CdS, gold nanoparticles, and an aptamer to create a photoelectrochemical aptasensor sensitive to the recognition of a widely used anti-inflammatory drug, diclofenac (Figure 7). The CdS incorporated into a graphene composite demonstrated excellent electron transfer properties. The detection limit obtained for this aptasensor was very low, at 0.78 nM, making it a promising tool for analyzing environmental samples [161].

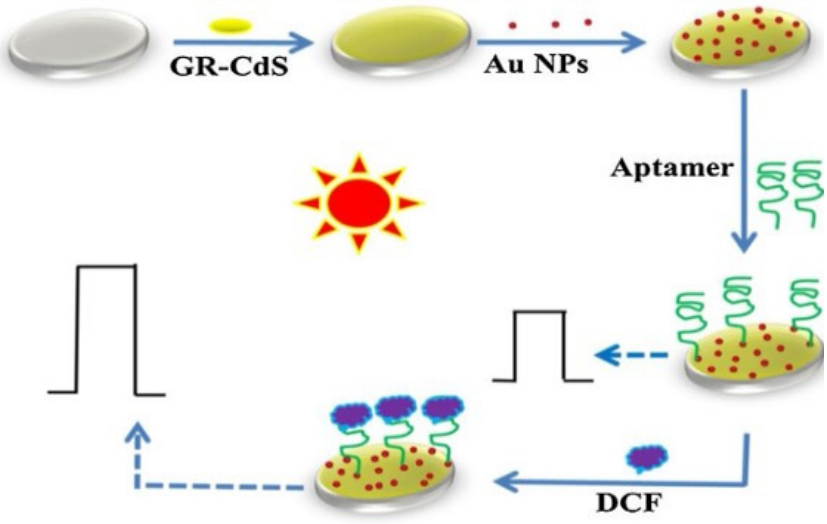


Figure 7. Schematic illustration of the fabrication process of aptasensor. (Reprinted with the permission of the publisher [161]).

4.3.3. Aptamers

In addition to antibiotics and hormones, antifungal agents such as metronidazole were also identified in environmental samples, specifically lake water. Zheng et al. used an ultrasensitive biosensor, GCE/PDDA-GN/DNA (a biosensor based on a glassy carbon electrode modified with poly(diallyldimethylammonium chloride)-functionalized graphene and DNA), to evaluate the amount of metronidazole in lakes, obtaining satisfactory results. Two electrochemical techniques, cyclic voltammetry and linear sweep voltammetry (LSV), led to the evaluation of this biosensor, which showed excellent stability, sensitivity, reproducibility, and a detection limit of 24 nM, demonstrating the enhanced electrocatalytic activity for the reduction of metronidazole. All these aspects illustrate the remarkable qualities of the biosensor, making it suitable for environmental applications in water quality assessment [162].

In recent years, biosensors such as the one developed by Hosu et al. [163] for detecting tetracycline in water samples have been developed. These were obtained by electrodepositing gold nanostructures on the surface of a glassy carbon electrode (GCE). To increase the sensitivity of the detection instrument, the authors immobilized DNA aptamers on the electrode surface with gold nanostructures using various thiol binding strategies. The resulting biosensors were tested on wastewater samples, offering researchers an accessible, time-efficient, and adaptable approach.

Table 3. Examples of biosensors developed for quantifying pharmaceutical pollutants in different water samples.

Biosensor	Method	Analyte Detected	LOD	Linear Range	Type of Water Sample	Ref.
Carbon Nanotubes						
GCE/Chit-IL-MWCNT/TPA/aptamer/MB	DPV	Ibuprofen	0.00002 µmol/L	0.00007–6 µmol/L	Wastewater	[164]
GCE/SWCNT-Cdot/laccase	DPV	17α-ethinylestradiol	0.004 µmol/L	0.05–7 µmol/L	Tap water	[160]
GCE/SWCNT-Chit/CTAB-AuNPs/antigen/Ab1/Ab2-AgNPs-DFNS	AMP	Sulfamethazine	0.00024 µmol/L	0.0016–0.15 µmol/L	River water Lake water Pond water	[165]
Graphene, metallic nanoparticles						
Au/MCH/graphene-aptamer	DPV	17β-estradiol	5.0×10^{-8} M	7.0×10^{-8}–1.0×10^{-5} M	Tap water	[85]
FTO/graphene-CdS/AuNPs/aptamer	PhotoAMP	Diclofenac	0.00078 µmol/L	0.001–0.15 µmol/L	Lake water	[161]
GCE/graphene-PDDA/DNA	LSV	Metronidazole	0.024 µmol/L	0.05–100 µmol/L	Lake water	[162]
Anti-sulfamethazine monoclonal antibody/Au NDs/GCE	LSV	Sulfamethazine	0.12 ng/mL	0.33–63.81 ng/mL	Environmental waters	[166]
Enzyme, antibody, and aptamer						
DNA aptasensor	SWV	Azlocillin	1.2 pg/mL	-	Waste water	[167]
Novel 3D honeycombed goldnanovoids@aptamer nanostructured platform with ferrocene labeling on aptamer strands	DPV	Tetracycline	1.2 nM	-	Wastewater	[163]
Molybdenum disulfide nanoparticles ($nMoS_2NPs$) deposited on ITO-coated glass substrate with Abs bonded through amide linkages	DPV	Ampicillin	0.028 µg mL^{-1}	-	Tap water	[168]
Aptasensor (Aptamer–complementary DNA (cDNA) oligonucleotides)	CV	Progesterone	0.9 µg·L^{-1}	10 µg·L^{-1}–60 µg·L^{-1}	Tap water	[169]
PEDOT (Poly (3,4-ethylenedioxy-thiophene))-GONs (graphene oxide nano-sheets)-Laccase/GCE	CV	Catechol	32 nM	-	Tap water	[170]
PEC (photoelectrochemical) Aptasensor NiFe LDH (layered double hydroxide)/g-CN (graphitic carbon nitride)/ITO electrode	LSV	Ofloxacine	0.035 pM	0.0001–1 nM	Tap and river water	[171]

4.4. Advantages and Limitations of Electrochemical Biosensors

Due to their simple chemical and physical design, portability, involvement of a small number of professionals in construction and analysis, low cost, and ease of handling, biosensors are explored in many interdisciplinary research areas and across various fields, such as environmental monitoring [172], agriculture [173], health [174], and chemistry [175], even leading to their large-scale production [176]. Electrochemical methods for the development and analysis of biosensors offer additional benefits, including rapid response, increased sensitivity and selectivity, high accuracy and precision, excellent detection limits, reproducibility, and the involvement of small sample volumes [177]. On the other hand, this widespread use requires testing and applying these devices on a wide range of sample types, including cell cultures, food supplements, biological fluids, and environmental samples (water and soil) [178]. Some inconveniences and limitations of electrochemical biosensors, such as limited lifespan or usability, increased sensitivity to interactions with various analytes, and a limited temperature range, have led researchers to improve their performance by using new modifying materials [177]. These limitations are further amplified by factors such as high salinity, pH fluctuations, variations in ionic concentration, and the presence of organic or inorganic interferences, which can significantly impact the accuracy and stability of sensor readings [179–181]. For example, in environments with highly acidic or alkaline pH levels, biosensors may experience changes in sensitivity or specificity, reducing their ability to detect pharmaceutical substances with precision [181–183]. Wei et al. [181] report that in the interaction of diclofenac with DNA immobilized on the electrode, the forced increase in ionic strength of the solution by adding NaCl did not significantly affect the voltammetric behavior of diclofenac at pH 7.0, indicating the predominance of intercalative interaction. On the other hand, at pH 3.0, an increase in anodic current was observed upon the addition of NaCl, suggesting the presence of electrostatic interactions between diclofenac and DNA. This electrostatic interaction arises due to diclofenac being positively charged and DNA being polyanionic. Additionally, the presence of organic contaminants in water samples can interfere with electrochemical reactions, thereby affecting the stability of the sensor signal. It is noteworthy that the number of studies reporting biosensors for detecting pharmaceutical substances in environmental water samples is significantly lower compared with electrochemical sensors, largely due to challenges related to the stability and specificity of biomolecules used in biosensors under varying environmental conditions. This limitation could be addressed through the development of biosensors based on molecularly imprinted polymers or stable aptamers, which maintain high affinity and optimal resistance in complex environments, thus enabling reliable detection of pharmaceutical contaminants.

5. Solutions and Future Research Perspectives

Future research can explore several directions to improve electrochemical sensors and biosensors. The development of electrodes modified with eco-friendly nanocomposites, using natural polymers such as starch or gelatin and nanoparticles obtained through sustainable methods, could enhance the active surface area and sensitivity of sensors, especially in the detection of antibiotics in complex samples such as wastewater. For example, Safwat et al. [184] developed three eco-friendly microfabricated solid copper sensors for measuring triclosan in water. The sensors were enhanced by incorporating multi-walled carbon nanotubes and using β-cyclodextrin as a selective ionophore, achieving detection limits of 9.87×10^{-9} M, 9.62×10^{-10} M, and 9.94×10^{-11} M, respectively. The use of nanocomposites made from natural materials such as chitosan or lignin to replace β-cyclodextrin could further improve the sensor's selectivity for triclosan in the presence of interfering compounds. For lignin, the pathway could involve extraction

from plant biomass followed by chemical modifications to improve its interaction with triclosan. Similarly, for chitosan, the synthesis could involve the deacetylation of chitin, resulting in a biocompatible ionophore with the potential for selective binding of triclosan. Performance comparisons could highlight how these alternatives compare to β-cyclodextrin in terms of sensitivity, detection limits, and interference resistance in sensor applications. Furthermore, designing multiplexed electrochemical platforms capable of simultaneously detecting multiple types of antibiotics will significantly improve the rapid and efficient monitoring of contaminants. Optimizing sensor structures to reduce detection limits and increase reproducibility is essential for analyzing trace pharmaceutical contaminants. A promising direction would be the integration of these sensors into portable devices or smart monitoring systems connected to the internet, which could collect real-time data and automatically analyze them using machine learning algorithms. AI systems could predict contamination levels based on historical data and optimize calibration and maintenance intervals, reducing operational costs.

At the same time, biosensors could integrate nanomaterials functionalized with specific biomolecules, such as monoclonal antibodies or aptamers, for the rapid and specific detection of pharmaceutical contaminants, pesticides, and pathogenic bacteria in water and food. Biosensor platforms could also use integrated microfluidics and signal amplification through the plasmonic effect of gold nanoparticles, enabling simultaneous analysis of multiple types of contaminants at extremely low concentrations. These biosensors could be connected to IoT systems for continuous monitoring of environmental media quality. These innovations would significantly expand the applicability of electrochemical sensors and biosensors in environmental monitoring and public health safety.

6. Conclusions

This review provides an overview of recent advancements in the development of electrochemical sensors and biosensors for detecting pharmaceutical contaminants in water samples, including surface water, wastewater, and bottled water. Recent studies highlight the importance of electrode-modifying materials, which have had a significant impact on sensor performance, improving sensitivity, selectivity, and detection limits. Pharmaceutical contaminants identified in the research include a variety of substances such as anti-inflammatory drugs (diclofenac, ibuprofen), hormones (17α-ethinylestradiol), antibiotics (sulfamethoxazole, ciprofloxacin), and others, emphasizing the complexity of the challenges faced in analyzing these compounds. Nanomaterials, including conductive polymers, graphene, and carbon nanotubes, have been essential in miniaturizing sensors and improving analytical methods, demonstrating increased reproducibility and precision. Future research in electrochemical sensors and biosensors may include the use of eco-friendly nanocomposites, the development of multiplexed platforms for simultaneous detection of pharmaceutical compounds, optimization of detector element structures for enhanced sensitivity, and integration into portable devices, thereby contributing to the effective in situ monitoring of contaminants and ensuring the safety of natural water sources.

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